

Physiology of the Nervous system

Lecture (4)



Sensory pathway 2. Pain. Lateral and Anterior spinothalamic tracts

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*Illustrations from (Lippincott illustrated medical
physiology)*



Pain and nociception..

- ❖ Nociception is the detection of noxious stimuli and the subsequent transmission to the CNS. Pain, is the product of higher cortical processing. Pain is affected by attention, autonomic variables and more.....

❖ Substances which cause pain:

- ❖ Serotonin, potassium, proteolytic enzymes, acetylcholine, bradykinin, etc.....



Pain components.....

Motor component

Withdraw
Muscle
guarding

Emotional component

Crying
Fear
Anxiety

Autonomic component

Tachycardia
Bradycardia
sweating



Pain...

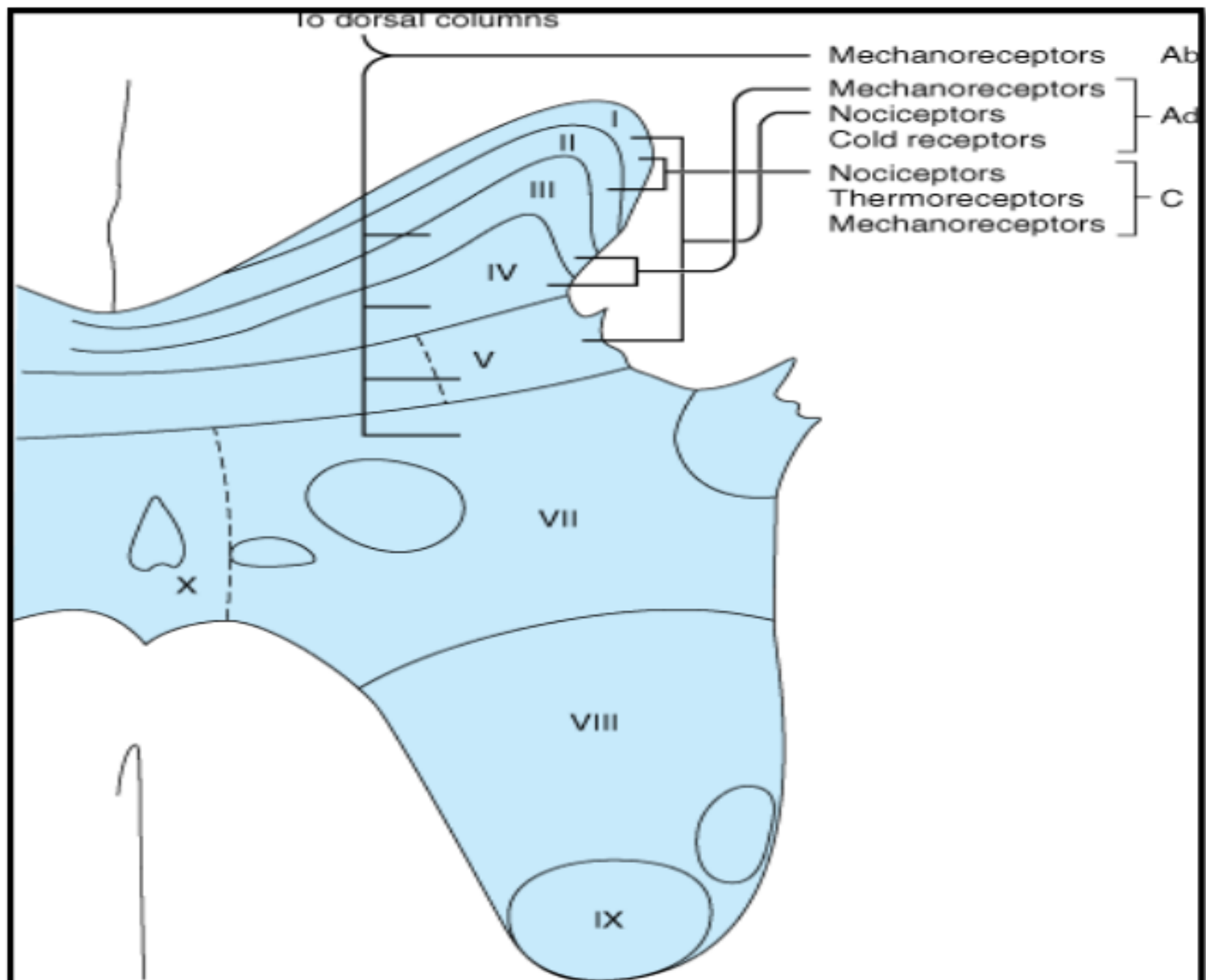
- (Qualities) Fast pain v. slow pain
- Deep pain v. superficial pain (cutaneous)
- Visceral from the inner organs (spasm or distension or ischemia,.....)
- Cutaneous is usually localized. Deep may refer to other areas and visceral is both localized and referred



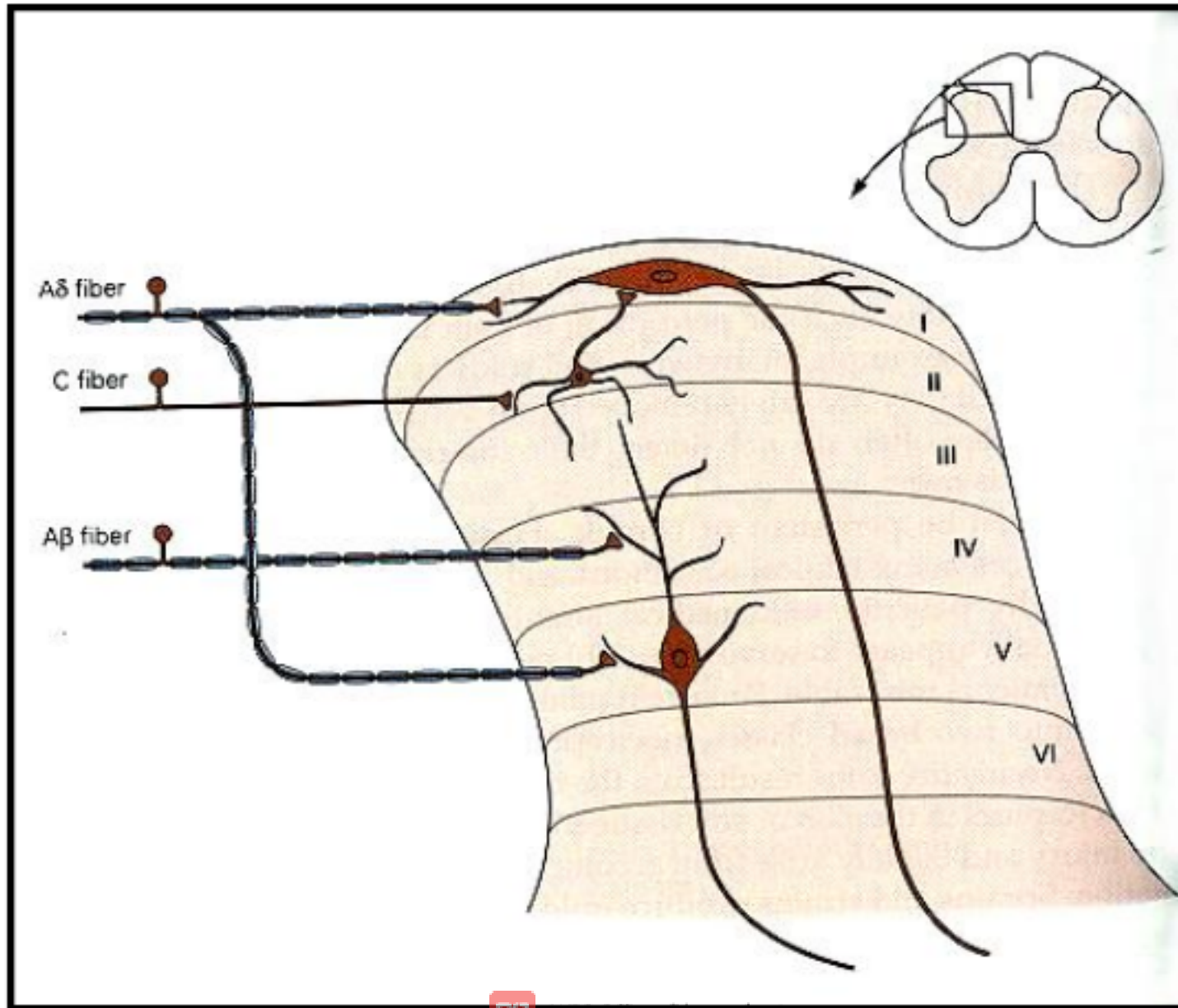
Pathway of pain:

- **Receptors**
 - free nerve endings .
- **Peripheral nerve :**
 - type of fibres : A delta, C fibres .
 - diameter : 2-5 micrometer, 0.4 -1.2 micrometer.
 - conduction velocity :12-30 m\s, 0.5 -2 m\s.





Connections in the dorsal horn



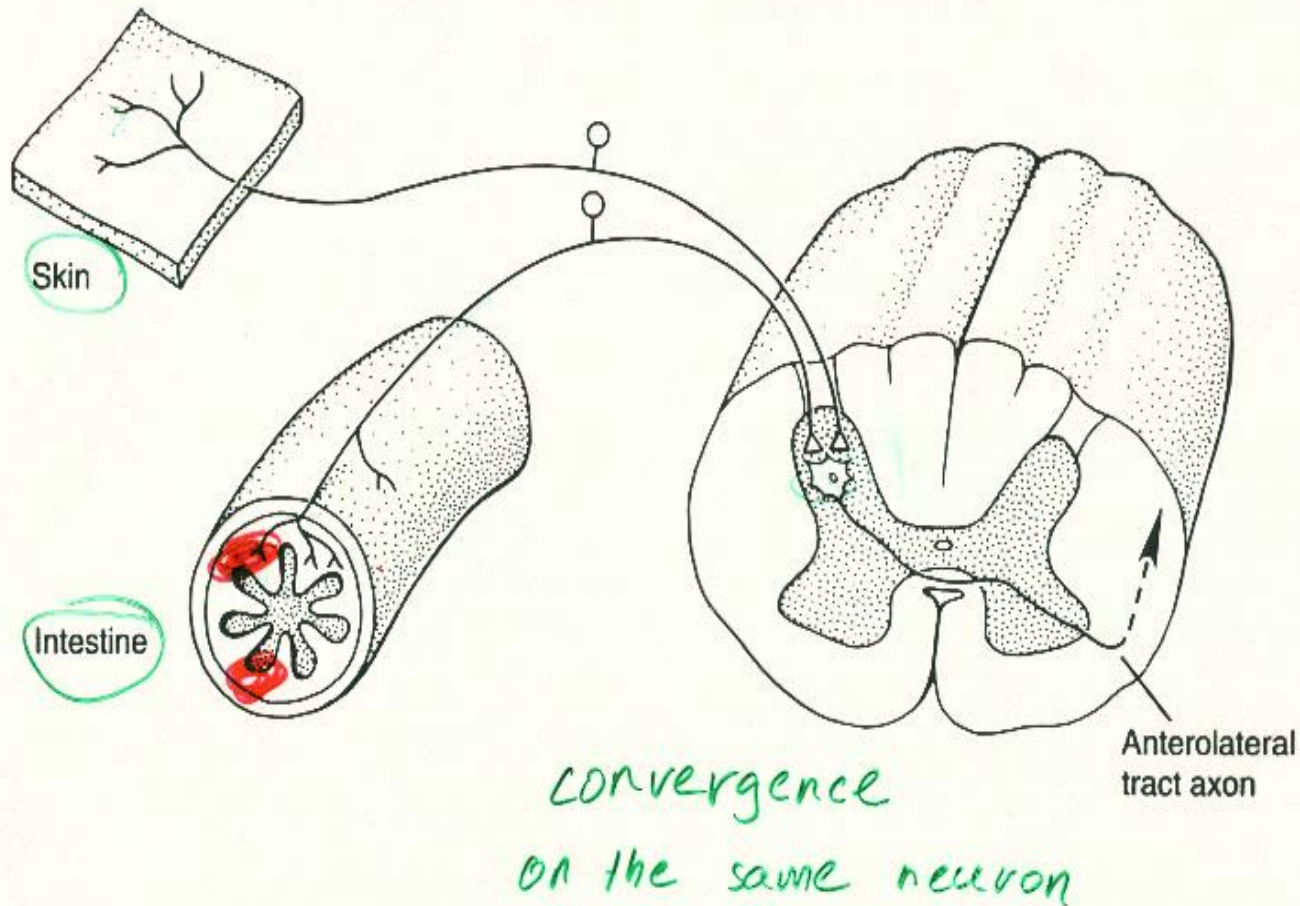
Theories of Referred pain...

- Convergence and facilitation

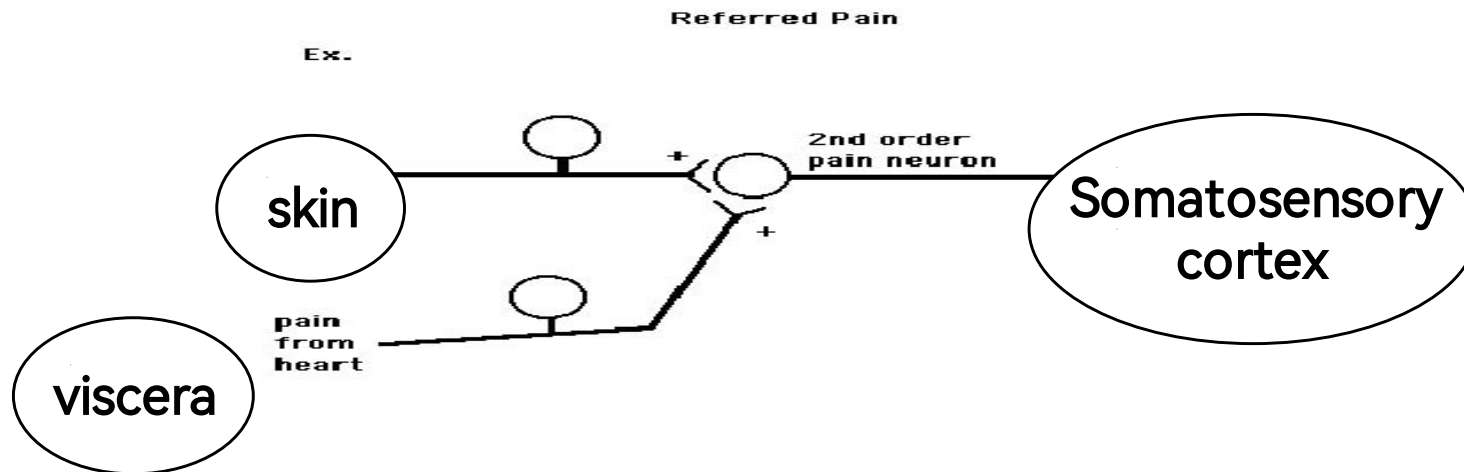


Convergence theory

referred pain: explanation 1.



Facilitation theory :



How to inhibit pain centrally:

- 1. Rubbing the area around an injury usually relieves pain
- 2. The Wide variation in the degree to which patients react to pain.
- 3. Ignorance of some injuries while playing a game or during a battle .
- 4. Irritation of the skin overlying a diseased viscus eg,
 - With mustard plaster, usually relieves pain .
- 5. Acupuncture .



The opiate system of the brain

- Naturally occurring opioids are mainly:

1.Enkephalins.

2.Endorphins.



Endorphins

- Mainly found in :
 1. hypothalamus .
 2. pituitary
- Importance :
 1. prevent withdrawal symptoms.
 2. role during parturition .



Stimuli for opioid release :

- Stress (Stress analgesia) .
- Sexual stimulation.
- Acupuncture.
- Placebos.



Gate control theory

- proposed by Melzack and Wall in 1965 .
- Briefly, it states that :
for nociceptive signals to enter the CNS, they have to pass through a gate in the spinal cord that can either be closed (inhibited) or opened (facilitated) .



Lateral spinothalamic tract (LSTT)

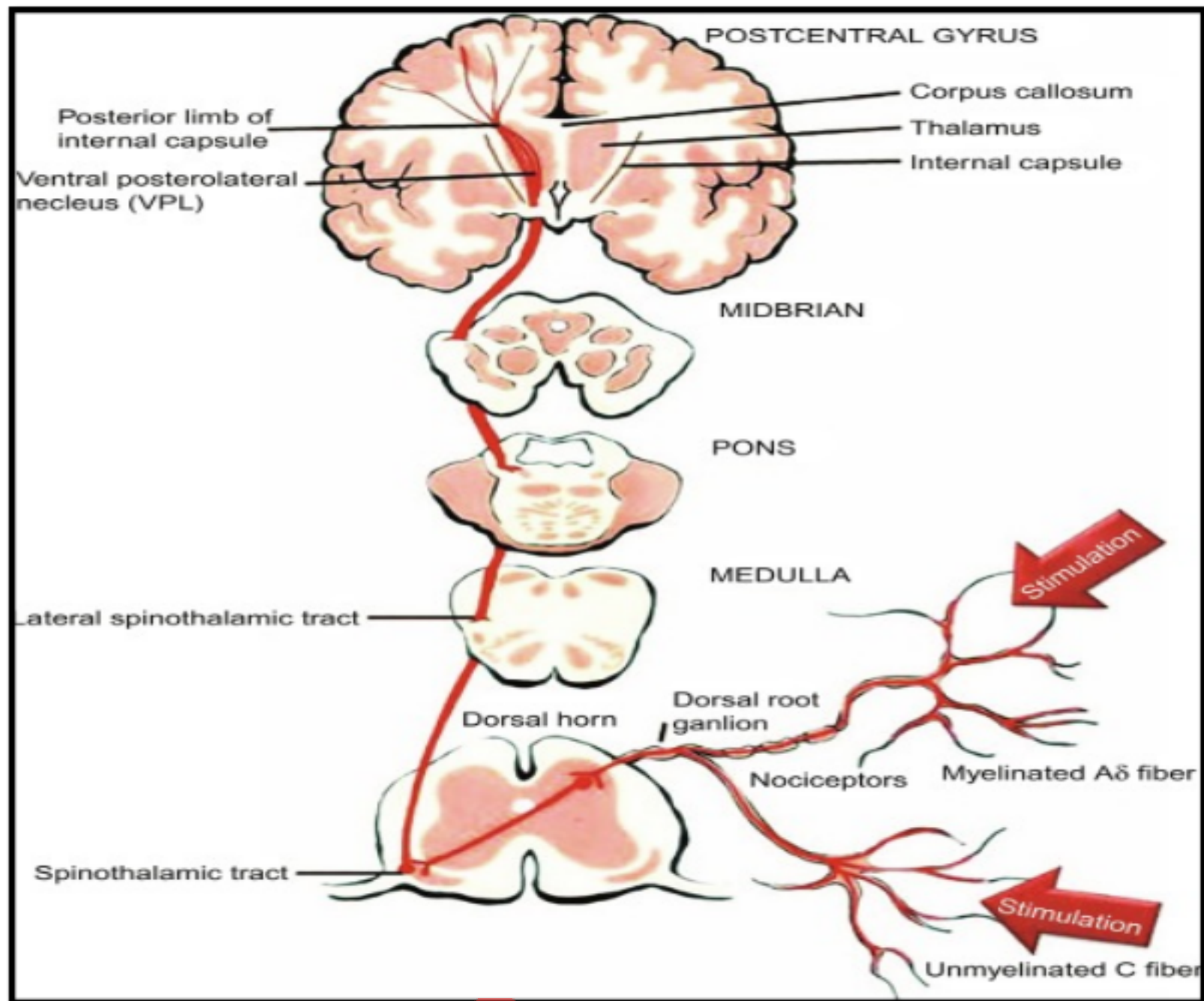
- **Receptors:** free nerve endings
- **1st order neuron:** posterior root ganglion
- **2d order neuron:** substantia gelatinase
- **3d order neuron:** ventral posterolateral nucleus of the thalamus
- **Cortex:** posterocentral gyrus



Anterior spinothalamic tract (ASTT)

- **Receptors:** Pacinian corpuscle
- **1st order neuron:** posterior root ganglion
- **2d order neuron:** substantia gelatinase
- **3d order neuron:** ventral posterolateral nucleus of the thalamus
- **Cortex:** posterocentral gyrus





Physiology lit. (recommended)...

- Guyton and Hall textbook of physiology
- **Lippincott illustrated physiology**
- Ganong review of medical physiology
- Linda Costanzo textbook of medical physiology
- BRS, Linda Costanzo
- Essential to physiology

